

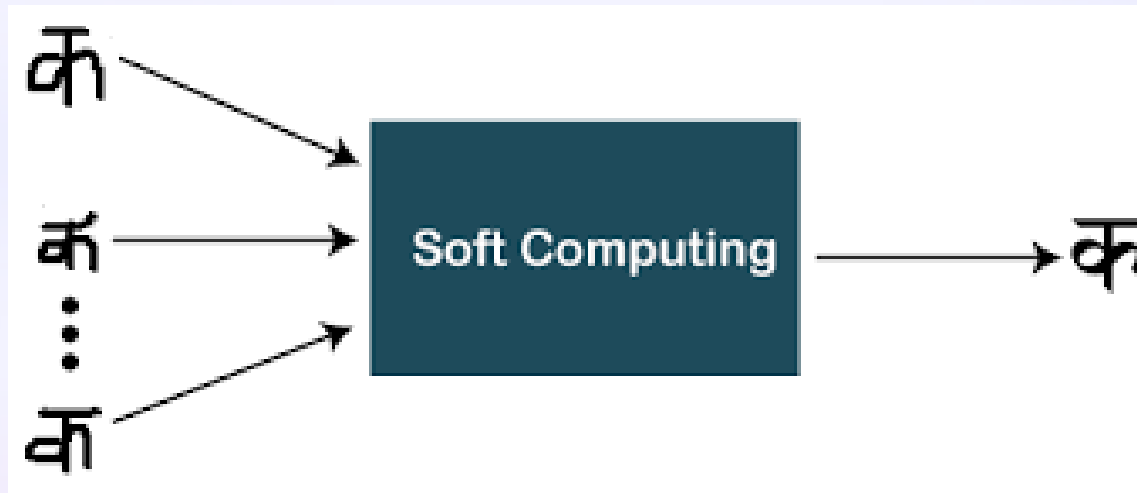
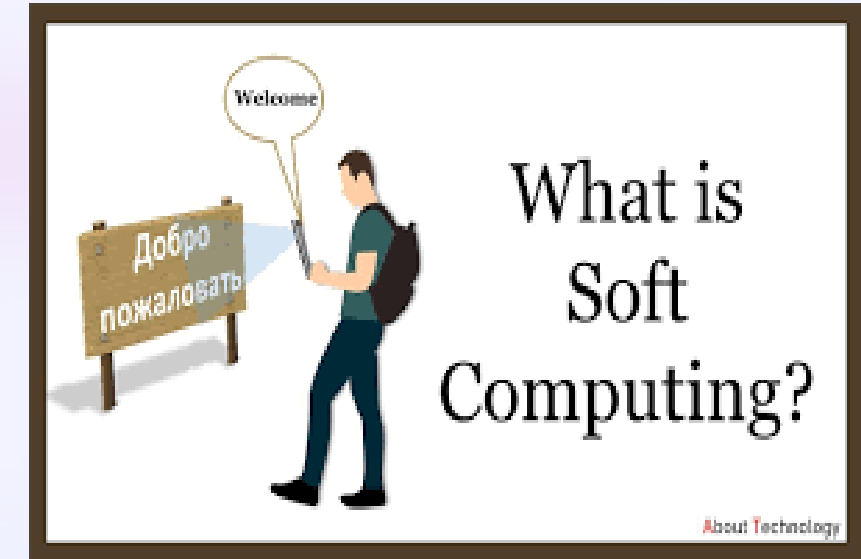
Introduction to Soft Computing

- NAME - ADITYA KUMAR
- UNIVERSITY ROLL NO :
34230821034
- PAPER NAME : Soft Computing
- PAPER CODE : PECAIML-603
- SEMESTER/YEAR : 6TH SEM/ 3RD YEAR
- DEPARTMENT : CSE(AI&ML)



What is Soft Computing?

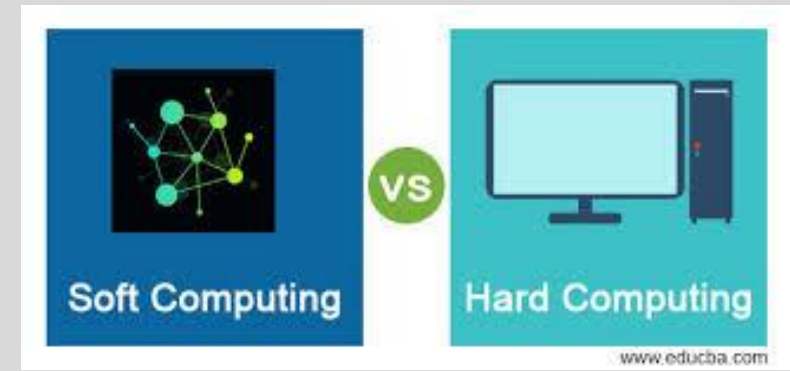
- 1) Soft computing is the reverse of hard (conventional) computing. It refers to a group of computational techniques that are based on [artificial intelligence \(AI\)](#) and natural selection.
- 2) It provides cost-effective solutions to the complex real-life problems for which hard computing solution does not exist.
- 3) Zadeh coined the term of soft computing in 1992. The objective of soft computing is to provide precise approximation and quick solutions for complex real-life problems.
- 4) In simple terms, you can understand soft computing - an emerging approach that gives the amazing ability of the human mind. It can map a human mind and the human mind is a role model for soft computing.



Need for Soft Computing

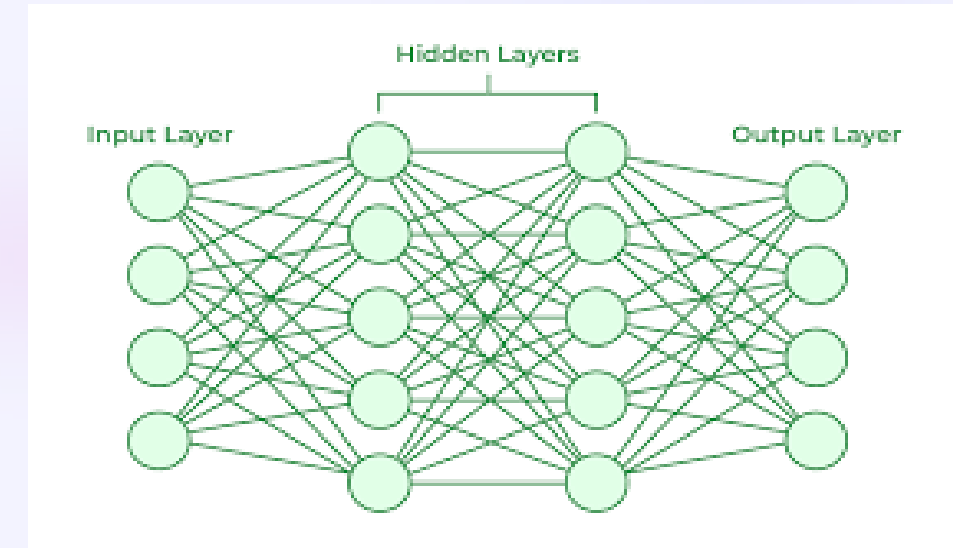
- 1) Sometimes, conventional computing or analytical models does not provide a solution to some real-world problems. In that case, we require other technique like soft computing to obtain an approximate solution.
- 2) Hard computing is used **for solving mathematical problems** that need a precise answer. It fails to provide solutions for some real-life problems. Thereby for real-life problems whose precise solution does not exist, soft computing helps.
- 3) When **conventional mathematical and analytical models** fail, soft computing helps, e.g., You can map even the human mind using soft computing.
- 4) Analytical models can be used for solving mathematical problems and valid for ideal cases. But the real-world problems do not have an ideal case; these exist in a non-ideal environment.
- 5) Soft computing is not only limited to theory; it also gives insights into real-life problems.
- 6) Like all the above reasons, Soft computing helps to map the human mind, which cannot be possible with conventional mathematical and analytical models.

SOFT COMPUTING VS HARD COMPUTING



Parameters	Soft Computing	Hard Computing
Computation time	Takes less computation time.	Takes more computation time.
Dependency	It depends on approximation and dispositional.	It is mainly based on binary logic and numerical systems.
Computation type	Parallel computation	Sequential computation
Result/Output	Approximate result	Exact and precise result
Example	Neural Networks, such as Madaline, Adaline, Art Networks.	Any numerical problem or traditional methods of solving using personal computers.

Basic Definition of Artificial Neural Networks (ANN)



Structure

An artificial neural network is composed of interconnected nodes, inspired by the biological neural networks in the human brain. These nodes, also called artificial neurons, process input data and produce output signals.

Learning Paradigms

ANNs are trained using various learning paradigms such as supervised learning, unsupervised learning, and reinforcement learning, enabling them to learn from input-output data patterns.

Applications

They find application in diverse domains including pattern recognition, classification, forecasting, optimization, and complex system control due to their ability to model complex non-linear relationships.

Basic Definition of Fuzzy Logic

① Handling Uncertainty

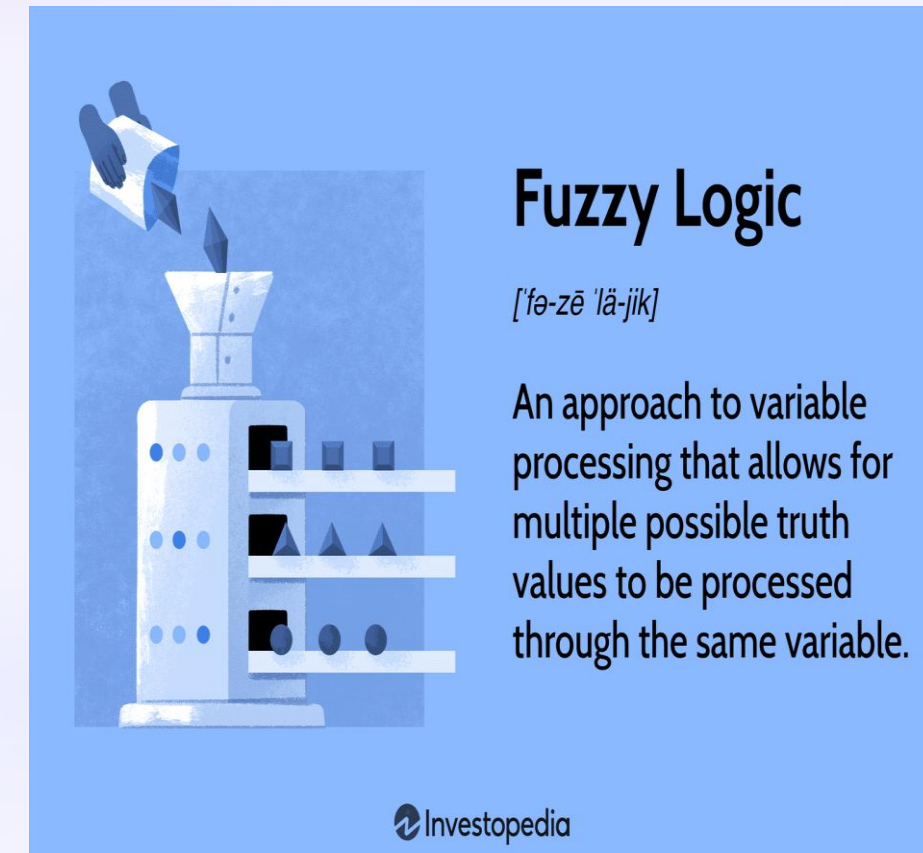
Fuzzy logic provides a way to represent and process approximate reasoning, allowing for the manipulation of concepts and knowledge that are not precise or clearly defined.

② Membership Functions

It employs membership functions to quantify the degree of truth as a continuous value between 0 and 1, enabling the representation of imprecise and vague linguistic terms.

③ Real-world Applications

Fuzzy logic is used in systems where linguistic terms and human-like reasoning are involved, such as in controlling home appliances, automobile subsystems, and industrial processes.



Basic Definition of Genetic Algorithms

1

Evolutionary Optimization

Genetic algorithms mimic the process of natural selection and evolution to search for optimal solutions to complex problems through genetic recombination and mutation.

2

Selection and Crossover

They involve the selection of parent solutions based on fitness and their recombination using crossover operations, producing new candidate solutions for the next generation.

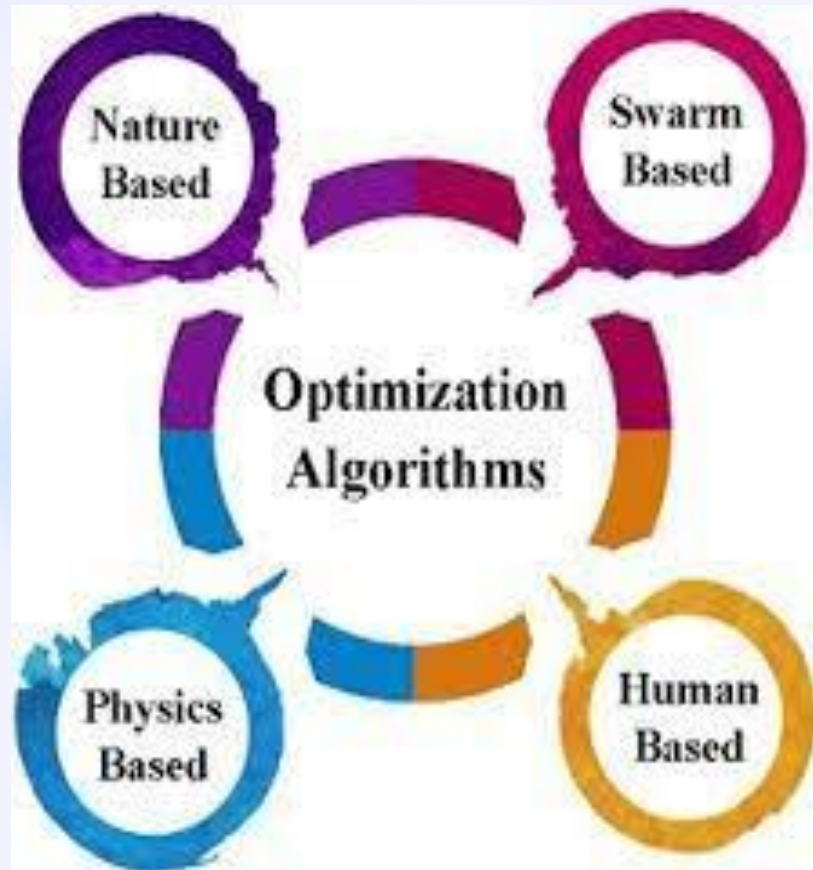
3

Application Range

These algorithms are widely applied in optimization problems, parameter tuning, robotics, and machine learning tasks, particularly when the search space is large and complex.



Basic Definition of Optimization



Objective Functions

Optimization involves defining an objective function that needs to be maximized or minimized based on specified constraints, providing an optimal solution for the problem at hand.

Search Space Exploration

Various optimization algorithms explore the search space systematically or heuristically to locate the best solution, considering the trade-offs between exploration and exploitation.

Real-world Implementation

Optimization techniques are prevalent in diverse fields including engineering design, logistics, finance, and operations research, aiding in decision-making and resource allocation.

Applications of Soft Computing

1

Pattern Recognition

Soft computing techniques are extensively used for recognizing and classifying patterns in fields such as image processing, speech recognition, and bioinformatics.

2

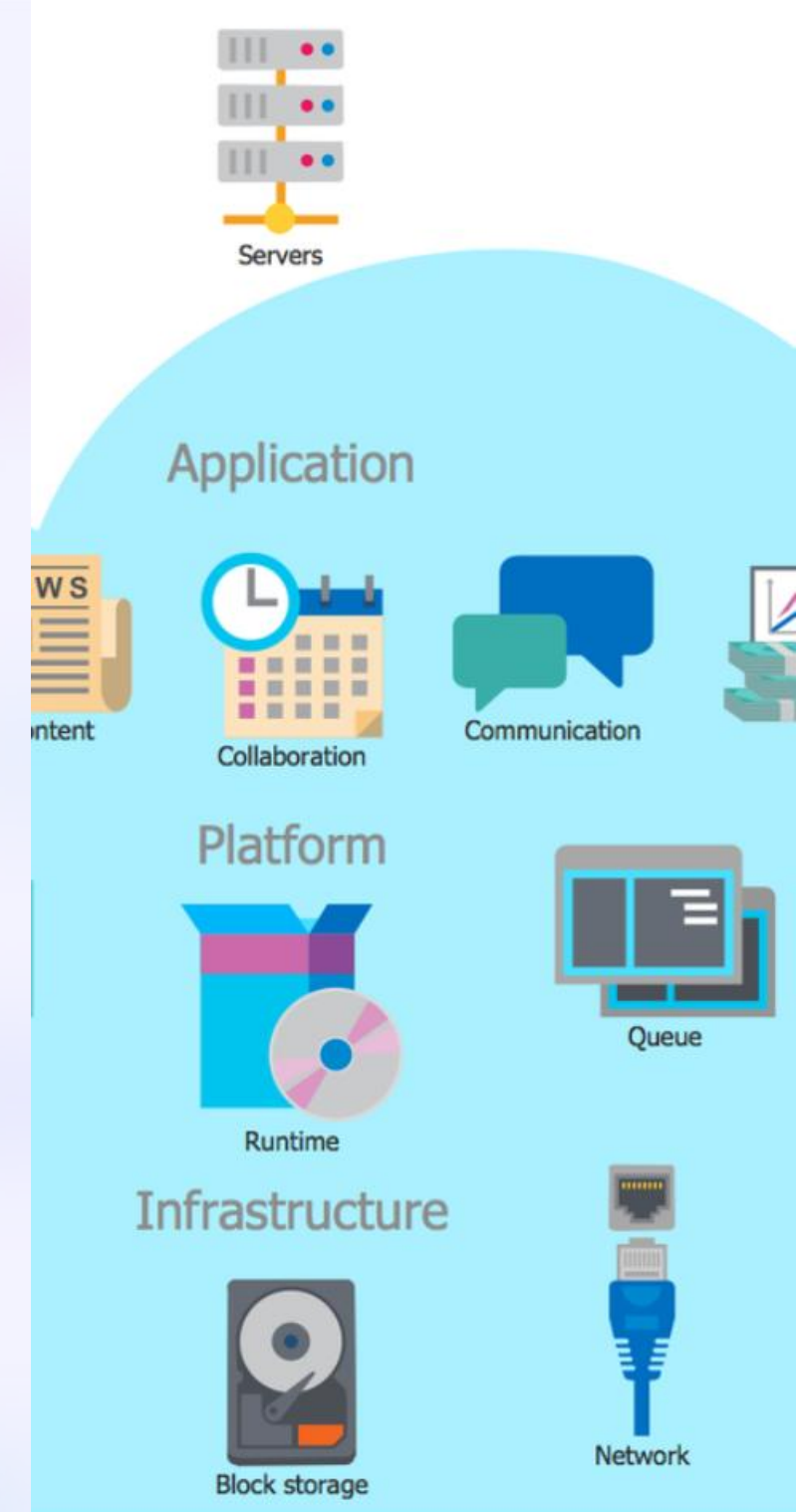
Data Mining

They play a vital role in extracting valuable insights from large datasets, aiding in various domains including marketing, healthcare, and financial analysis.

3

Autonomous Systems

Soft computing methods contribute to the development of autonomous systems capable of making decisions and adapting to dynamic environments without human intervention.



Advantages/Applications of Soft Computing

- 1) **Robustness:** Soft computing techniques are robust and can handle uncertainty, imprecision, and noise in data, making them ideal for solving real-world problems.
- 2) **Approximate solutions:** Soft computing techniques can provide approximate solutions to complex problems that are difficult or impossible to solve exactly.
- 3) **Non-linear problems:** Soft computing techniques such as fuzzy logic and neural networks can handle non-linear problems effectively.
- 4) **Human-like reasoning:** Soft computing techniques are designed to mimic human-like reasoning, which is often more effective in solving complex problems.
- 5) **Real-time applications:** Soft computing techniques can provide real-time solutions to complex problems, making them ideal for use in real-time applications.

There are several applications of soft computing where it is used. Some of them are listed below:

- 1) It is widely used in **gaming products like Poker and Checker.**
 - 2) In kitchen appliances, such as **Microwave and Rice cooker.**
 - 3) In most used home appliances - **Washing Machine, Heater, Refrigerator, and AC** as well.
 - 4) Apart from all these usages, it is also used in **Robotics work** (Emotional per Robot form).
 - 5) **Image processing and Data compression** are also popular applications of soft computing.
 - 6) Used for handwriting recognition.
- As we already said that, soft computing provides the solution to real-time problems and here you can see that. Besides these applications, there are many other applications of soft computing.

Conclusion & Summary

Soft computing, with its diverse subfields such as artificial neural networks, fuzzy logic, and genetic algorithms, offers a powerful paradigm for handling complex, uncertain, and dynamic problems. As technology continues to advance, soft computing methods are expected to play an increasingly pivotal role in various applications ranging from intelligent systems to decision support systems.



THANK YOU